SPATIO-TEMPORAL ANALYSIS OF COASTAL CHANGE: USING REMOTE SENSING AND GIS FOR SHORELINE MAPPING AND BATHYMETRIC ASSESSMENT ALONG THE ALGERIA COASTLINE

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Chapter 1 — Introduction

1.1 Background

Coastal environments represent some of the most complex and rapidly changing landscapes on Earth. They are shaped by the interaction of oceanographic, geomorphological, and atmospheric processes, which together determine patterns of erosion, deposition, and sediment transport. The coast is not only a physical boundary but also an ecological hotspot, supporting mangroves, seagrass beds, coral reefs, sandy beaches, and wetlands. These habitats sustain diverse biological communities while providing crucial ecosystem services such as coastal protection, carbon sequestration, nutrient cycling, and fishery resources.

From a human perspective, coasts are focal points of settlement and development. Globally, more than 40% of the population lives within 100 km of the shoreline, and this figure is projected to increase with urbanization and economic growth. Ports, shipping lanes, oil and gas facilities, and tourism infrastructure all depend on coastal accessibility. This concentration of population and investment makes coasts highly vulnerable to natural hazards such as storm surges, flooding, sea-level rise, and tsunamis. At the same time, anthropogenic pressures like land reclamation, dredging, sand mining, and industrial discharge contribute to the degradation of coastal ecosystems.

Monitoring these environments requires accurate and consistent data. Traditional ground surveys of the seabed and shoreline, although precise, are often time-consuming, costly, and spatially limited. The development of Remote Sensing (RS) and Geographic Information Systems (GIS) provides a modern solution, enabling large-scale, cost-effective, and repeatable monitoring of coastal systems. RS and GIS allow the integration of bathymetric data with satellite imagery, providing tools to analyze both seafloor morphology and shoreline dynamics. Bathymetric information is essential for understanding sediment pathways, identifying slope variations, and delineating potential habitats, while shoreline mapping reveals trends of erosion and accretion that directly impact human settlements and ecosystems. Together, these tools form the basis of spatially informed coastal management.

1.2 Problem Statement

Angola's coastline extends for approximately 1,650 km along the South Atlantic Ocean, encompassing a range of geomorphological and ecological features. It is shaped by the influence of the Benguela Current system, which drives strong upwelling along the southern sector, as well as by fluvial inputs from major rivers such as the Congo, Kwanza, and Cunene. These natural processes create a mosaic of coastal environments, from sandy beaches and lagoons in the north to rocky shores and semi-arid deserts in the south. Despite this diversity, Angola's coastline is under increasing stress.

Rapid urbanization, particularly around Luanda and other major cities, has led to unregulated construction along vulnerable coastal strips. Oil exploration and offshore drilling contribute to habitat disturbance and pollution. Coastal erosion is evident in several locations, exacerbated by rising sea levels and intensifying storm surges linked to climate change. In addition, industrial dredging, sand mining, and harbor expansion have altered sediment balances, accelerating shoreline retreat in some areas while causing unnatural accretion in others.

Despite these challenges, there is limited integration of geospatial datasets that capture both onshore and offshore dynamics. Most existing studies focus either on ecological surveys of specific habitats or engineering assessments of shoreline protection projects. Bathymetric data from sources such as the General Bathymetric Chart of the Oceans (GEBCO) has not been fully utilized for combined analysis with shoreline mapping in Angola. Without such integrated analysis, decision-makers face knowledge gaps in identifying erosion-prone areas, assessing slope variations, and designing sustainable adaptation strategies. This study addresses the gap by using RS and GIS tools to combine bathymetric mapping with shoreline change detection.

1.3 Aim and Objectives

The overall aim of this study is to carry out a spatio-temporal analysis of Angola's coastline, integrating bathymetric and shoreline data using Remote Sensing and GIS techniques to assess coastal change and identify geomorphological features.

The specific objectives are:

1. To acquire and preprocess spatial datasets, including GEBCO bathymetric data, GRID3 Africa boundary shapefiles, and satellite imagery suitable for shoreline mapping.
2. To generate bathymetric products such as hillshade maps, depth transects, and 3D visualizations that reveal underwater slope variations and seafloor morphology.
3. To identify and classify major coastal features and potential habitat zones based on bathymetric and geomorphological patterns.
4. To detect shoreline change through the application of the Digital Shoreline Analysis System (DSAS) on multi-temporal satellite data.
5. To interpret the results within the context of coastal management and provide recommendations for mitigating erosion, conserving habitats, and planning sustainable development.

1.4 Study Area Description

The study area covers the entire Angolan coastline, which extends from the mouth of the Congo River in the north (near the border with the Republic of Congo) to the Cunene River in the south (bordering

Namibia). Geographically, it lies between latitudes 4°22′S and 18°S and is divided into northern, central, and southern coastal sectors, each with distinct environmental characteristics.

- **Northern Coast**: This sector is humid and influenced by river discharges, supporting estuaries, lagoons, and mangroves. Major rivers like the Congo and Kwanza bring significant sediment loads, shaping deltaic features and feeding nearshore sediment transport systems.
- **Central Coast**: Home to the capital city Luanda, this area is heavily urbanized and industrialized, with major ports, oil terminals, and growing metropolitan expansion. It experiences a mix of sandy beaches and rocky outcrops, but erosion and flooding are major concerns due to unplanned development.
- **Southern Coast**: Characterized by arid and semi-arid conditions, this sector is strongly influenced by the Benguela Current upwelling system. The seafloor bathymetry shows steep slopes and deeper waters closer to the coast. The semi-arid Namibe Desert reaches the shoreline here, creating a unique landscape.

Climatically, Angola's coast transitions from tropical humid in the north to semi-arid in the south. Rainfall decreases from over 1,200 mm annually in the north to less than 300 mm in the extreme south. This variation in climate influences river discharge, sediment transport, and coastal erosion dynamics. Economically, the coastline is of immense importance: it hosts Angola's oil production hubs, supports artisanal and commercial fisheries, and provides opportunities for tourism. However, its ecological value is equally significant, with critical habitats such as mangrove forests, turtle nesting beaches, and seabird colonies under threat from human and natural pressures.

1.5 Study Area Map

A study area map was developed to show the geographical extent of the Angola coastline. Administrative boundaries were obtained from the GRID3 Africa shapefile

The map is essential for situating the reader and linking spatial products with specific coastal locations. By including cartographic elements such as a scale bar, north arrow, and legend, the map allows replication of the study and comparison with other regional datasets. It also serves as the baseline for interpreting subsequent results such as hillshade maps, depth profiles, and shoreline change detection outputs.

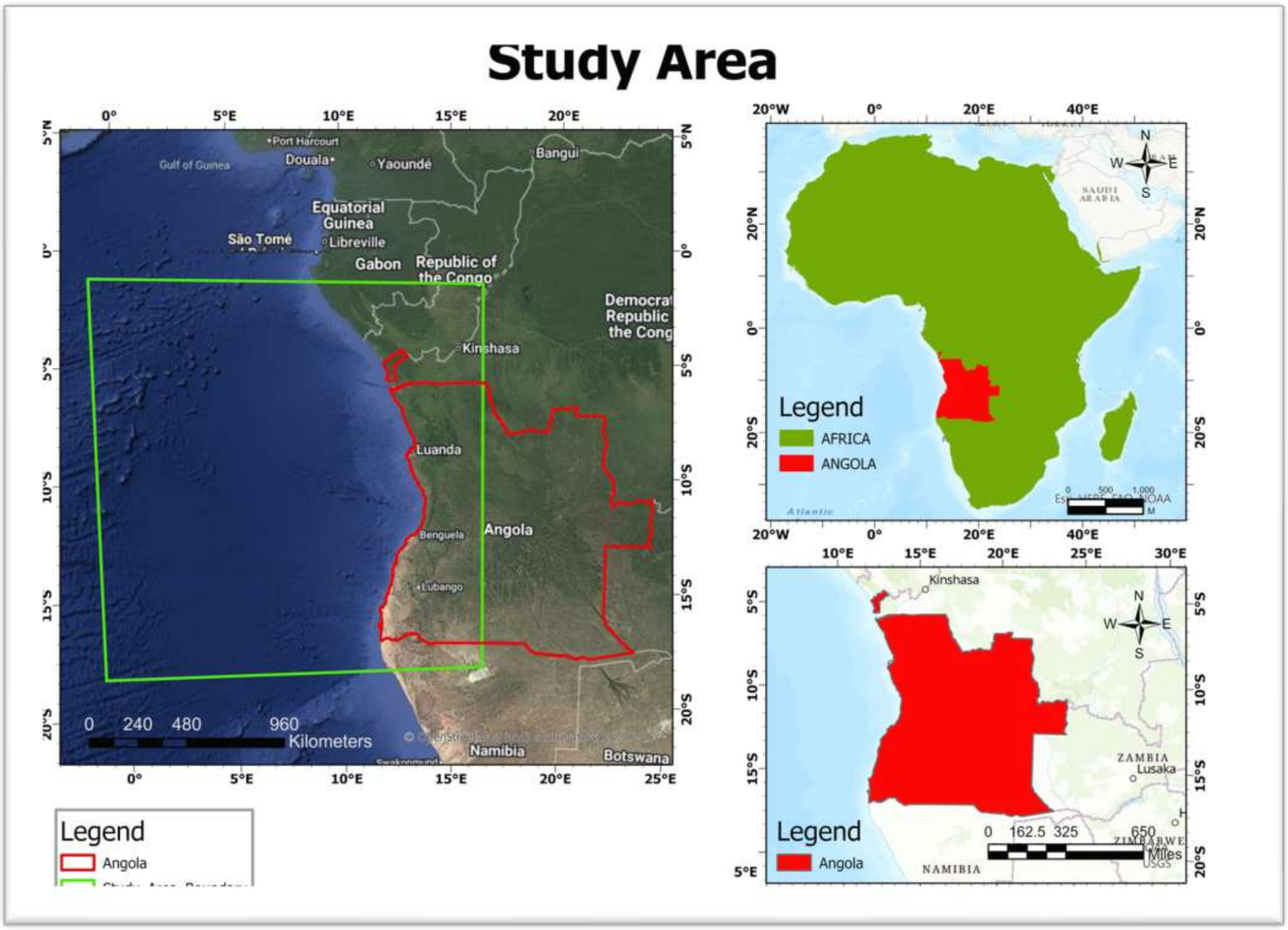


Figure 1.1 — Study area map of the Angola coastline showing administrative boundaries.

Chapter 2 — Literature Review

2.1 Coastal Geomorphology and Bathymetry

Coastal geomorphology focuses on the processes and landforms shaping the dynamic interface between land and sea. According to Kay and Alder (2017), coastal landscapes are influenced by multiple interacting factors, including tides, waves, currents, sediment supply, and sea-level fluctuations. These processes continually reshape coastal landforms such as beaches, dunes, estuaries, and deltas, making them highly vulnerable to both natural hazards and human activities. Nicholls and Cazenave (2010) emphasized that sea-level rise, when combined with local geomorphological conditions, can accelerate shoreline retreat and increase exposure to flooding.

Bathymetry provides insights into the depth and topography of the seafloor, which are critical for understanding coastal morphology and processes. Luijendijk et al. (2018) highlighted that variations in seafloor slope directly affect sediment transport pathways and wave energy dissipation, influencing whether coastlines experience erosion or accretion. In the case of Angola, Sousa, Pereira, and Dias (2017) noted that the continental shelf and the Benguela Current upwelling system strongly shape the coastal hydrodynamics, driving nutrient circulation and impacting both geomorphological and ecological dynamics.

Integrating bathymetric data with shoreline studies provides a more complete perspective of coastal systems. Ndour et al. (2018), in their study of West African coasts, demonstrated how combining shoreline change analysis with seafloor mapping improves predictions of erosion hotspots and habitat vulnerability. This underscores the importance of linking surface geomorphology with subsurface bathymetry for effective coastal management.

2.2 Remote Sensing and GIS Applications

The use of Remote Sensing (RS) and Geographic Information Systems (GIS) has transformed the study of coastal environments over the last three decades. Angwenyi and Nyadawa (2019) reviewed the role of RS and GIS in shoreline change detection and concluded that satellite imagery provides invaluable temporal records of coastal change at local to regional scales. These tools enable scientists to observe both short-term storm impacts and long-term erosion or accretion trends.

Bathymetric products such as the GEBCO global dataset (GEBCO Compilation Group, 2023) provide gridded seafloor depth data that, when integrated into GIS platforms, allow researchers to generate hillshade maps, slope models, and 3D seafloor visualizations. When combined with satellite imagery, these datasets offer powerful tools for assessing the interaction between land and sea. GRID3 (2022) shapefiles, for instance, support the integration of administrative and physical boundaries, ensuring that coastal assessments can be contextualized in relation to human activities.

The Digital Shoreline Analysis System (DSAS), developed by Thieler, Himmelstoss, Zichichi, and Ergul (2009), has become one of the most widely applied tools for quantifying shoreline change. It uses historical and modern shoreline datasets to compute statistical rates of erosion and accretion. Hapke et al. (2010) applied DSAS along the U.S. Atlantic coast, showing its effectiveness in revealing long-term shoreline dynamics, while Ndour et al. (2018) used similar approaches in Senegal and Benin, demonstrating its applicability in African contexts. Despite these advancements, applications of DSAS remain limited in Angola, where most studies have focused on oceanographic monitoring or resource exploitation rather than integrated shoreline and bathymetric assessments.

2.3 Research Gap

Although global studies have demonstrated the importance of integrating shoreline change detection with bathymetric mapping (Anthony et al., 2019; Luijendijk et al., 2018), such approaches remain underdeveloped in Angola. Previous work has largely concentrated on localized environmental issues such as fisheries, pollution, or sectoral development projects (UNEP, 2016). Sousa et al. (2017) highlighted the role of the Benguela Current in Angola's coastal processes, but their focus was primarily oceanographic rather than geomorphological.

The lack of comprehensive spatial studies limits the ability of decision-makers to anticipate erosion-prone zones, identify habitat vulnerabilities, and plan adaptation strategies. Furthermore, data scarcity, inconsistent monitoring programs, and restricted access to high-resolution imagery have constrained coastal research in Angola (UNEP, 2016). By leveraging open-source datasets such as GEBCO (2023) and GRID3 (2022), alongside multi-temporal satellite imagery, this study addresses these challenges. It integrates shoreline change detection with bathymetric analysis using GIS and DSAS to provide a framework for assessing coastal dynamics in Angola.

Chapter 3: Data and Methods

3.1 Data Sources

This study made use of three major datasets: bathymetric data, boundary shapefiles, and satellite imagery. Each played a different role in understanding Angola's coastline.

3.1.1 GEBCO Bathymetric Data

The main dataset used for seafloor analysis was the General Bathymetric Chart of the Oceans (GEBCO) 2023 Grid. This global terrain model provides bathymetric information at a resolution of 15 arc-seconds (about 500 m). The data was downloaded in **NetCDF format** and later converted into a GIS-compatible format (GeoTIFF). GEBCO was chosen because it offers consistent, high-quality data that can be used for generating different products such as hillshade maps, depth profiles, and 3D models of the seabed.

3.1.2 GRID3 Africa Boundary Shapefile

Administrative boundary data for Angola was obtained from GRID3 (Geo-Referenced Infrastructure and Demographic Data for Development). This shapefile was used to clip the bathymetric data and restrict the study area to Angola's coastal zone. Without this step, the analysis would include areas outside Angola, which were not relevant to this study.

3.1.3 Satellite Imagery for Shoreline Change

To study shoreline changes over time, satellite images were required. Landsat (30 m resolution) and Sentinel-2 (10 m resolution) images were downloaded for different years: **2000, 2010, 2020, and 2023**.

These images were obtained from the **USGS Earth Explorer** and the **Copernicus Open Access Hub**.

By comparing images from different time periods, the study was able to track changes in Angola's coastline.

3.2 Data Preprocessing

Before analysis, all datasets were prepared to ensure accuracy and compatibility:

- The **GEBCO bathymetric data** (GeoTIFF) format using ArcGIS.
- The **Angola boundary shapefile** was used to clip the bathymetric grid, leaving only the portion that covers Angola's coastal waters.
- **Satellite images** were corrected for atmospheric effects and processed using the Semi-Automatic Classification Plugin (SCP) in ArcGIS. Cloud-free images were selected where possible to ensure clearer shoreline boundaries.
- False color composites were generated (bands 4-3-2 for Landsat, bands 8-4-3 for Sentinel-2) to make the difference between land and water more visible.
- All datasets were projected into **WGS 1984 UTM Zone 33S**, the standard projection for Angola, to keep all spatial data consistent.

3.3 Bathymetric Mapping

The bathymetric dataset was used to generate several outputs that describe the underwater topography of Angola's coast.

3.3.1 Hillshade Map

A hillshade map was created from the bathymetric grid. This map simulates how light and shadow fall on the seabed, making ridges, valleys, slopes, and depressions easier to see. It provides an overall visual impression of seafloor morphology.

3.3.2 Depth Profiles (A–H)

Eight transects (A–H) were drawn perpendicular to the coast at evenly spaced intervals. Depth values were extracted along each transect, producing profiles that show how the seafloor depth increases from the shoreline outwards. These profiles help identify areas where the slope is gentle or steep, which may influence erosion, sediment transport, and possible human activities like fishing or dredging.

3.3.3 Composite Depth Profile

All eight depth profiles were combined into a single composite graph. This allowed comparison between different sections of the coast, highlighting areas where the seabed is shallow versus areas where it deepens quickly.

3.3.4 3D Visualization

Using QGIS 3D view and ArcGIS ArcScene, the bathymetric data was extruded to create a 3D model of the seafloor. This visualization provided a more realistic perspective, showing depth variation, slope breaks, and possible submarine features such as ridges or troughs.

3.4 Shoreline Analysis with DSAS

The **Digital Shoreline Analysis System (DSAS v5.1)** extension in ArcGIS was used to calculate rates of shoreline change:

- First, shorelines were digitized from the processed Landsat and Sentinel images using the **Normalized Difference Water Index (NDWI)** method, which separates land and water boundaries.
- A baseline was drawn parallel to the shoreline.
- DSAS then cast transects at 200 m spacing across the coast.
- For each transect, DSAS calculated rates of change using:
    - **End Point Rate (EPR):** change between the earliest and most recent shoreline positions.
    - **Linear Regression Rate (LRR):** a statistical estimate of long-term change across all shoreline years.
- Results showed whether sections of the Angola coastline were experiencing erosion (shoreline retreat) or accretion (shoreline advance).

3.5 Identification of Coastal Features and Habitat Zones

Using both bathymetric products and shoreline maps, coastal features and habitats were identified. This included:

- **Slope variations** (steep underwater cliffs vs. gently sloping seabeds).
- **Habitat zones**, such as mangrove swamps, tidal flats, nearshore sandbars, and estuaries.
- **Potential biodiversity hotspots**, where seabed morphology creates conditions for marine habitats.

These interpretations are essential because bathymetry not only shows physical depth but also indicates areas suitable for different ecological systems.

3.6 Workflow Summary

The methodology followed a structured approach:

1. **Data acquisition** from GEBCO, GRID3, USGS, and Copernicus.
2. **Data preprocessing** including clipping, reprojection, and atmospheric correction.
3. **Bathymetric mapping**, creating hillshade maps, transect profiles, and 3D visualizations.
4. **Shoreline extraction and change analysis** using DSAS.
5. **Identification of coastal features and habitat zones** for environmental interpretation.
6. **Integration of results** into maps, graphs, and visualizations for final reporting.

CHAPTER 4: RESULTS AND DISCUSSION

4.1 Hillshade Map of the Seafloor

The hillshade map of Angola's coastal seabed (Figure 4.1) provided a first-level visualization of underwater relief and morphology. By simulating the effect of light and shadow across the seafloor, the hillshade product brought out subtle differences in slope and elevation that are not immediately visible from raw bathymetric data.

- **Northern Angola:** The hillshade revealed a broad continental shelf with smooth gradients, indicating a gently sloping seabed. This morphology favors sediment deposition and supports extensive shallow-water habitats. Such areas are crucial for artisanal fisheries, seagrass meadows, and possible coral presence.
- **Central Angola:** In contrast, the central section showed steeply sloping seabeds where the depth drops sharply within a short distance from the coast. This suggests the presence of submarine canyons or fault-controlled escarpments. These features play a major role in sediment transport offshore and may also act as conduits for nutrient upwelling.
- **Southern Angola:** The southern section displayed mixed morphologies, including broad, flat terraces interrupted by slope breaks. This region is influenced strongly by the Benguela Current, which redistributes sediments and shapes the seabed into sandbanks and ridges.

Overall, the hillshade product was essential for identifying spatial variability in slope gradients, and it formed the baseline for interpreting subsequent bathymetric analyses.

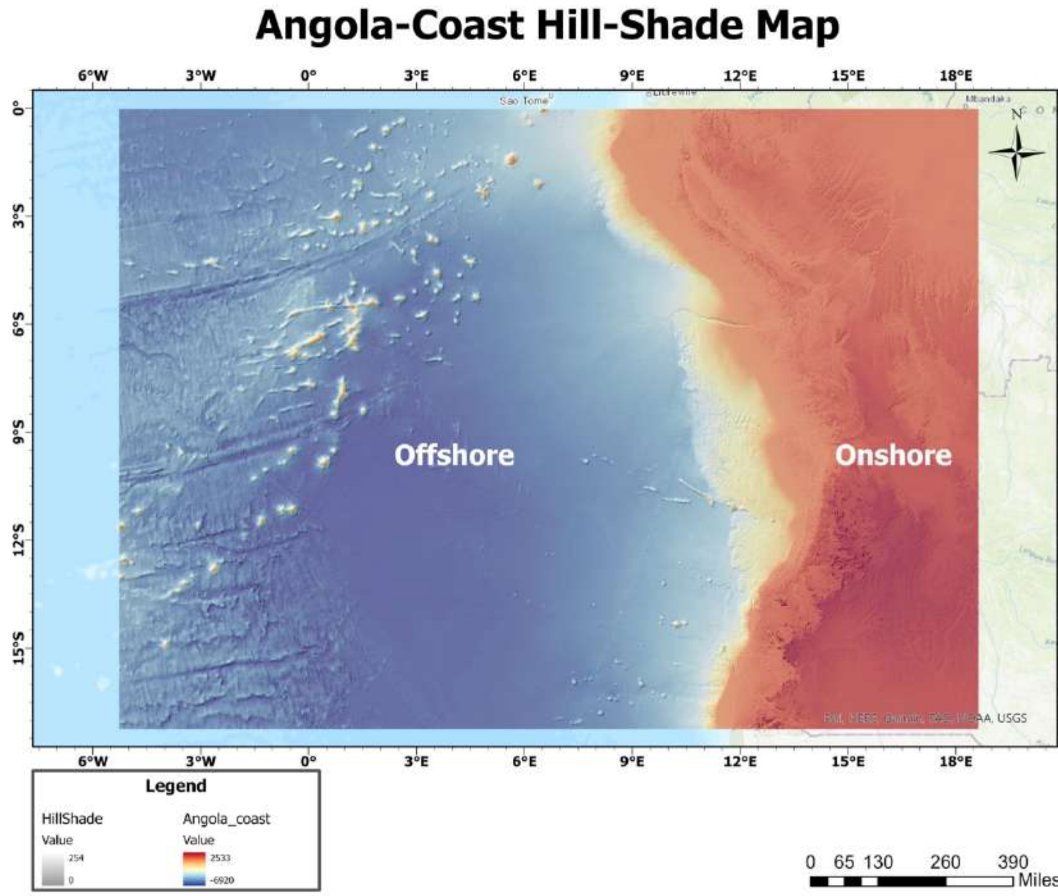


Figure 4.1: Hillshade Map of Angola Seafloor.

4.2 Transect View Map

A transect view (Figure 4.2) was prepared to show the exact positions of the eight depth profiles (A–H). Each transect was drawn perpendicular to the coastline, starting from the shoreline and extending offshore into deeper waters. The transects were evenly spaced to ensure a representative coverage of the Angolan coastline.

This layout allowed for the comparison of slope characteristics across northern, central, and southern sectors of the coast. It also provided a systematic framework for analyzing spatial differences in seabed topography.

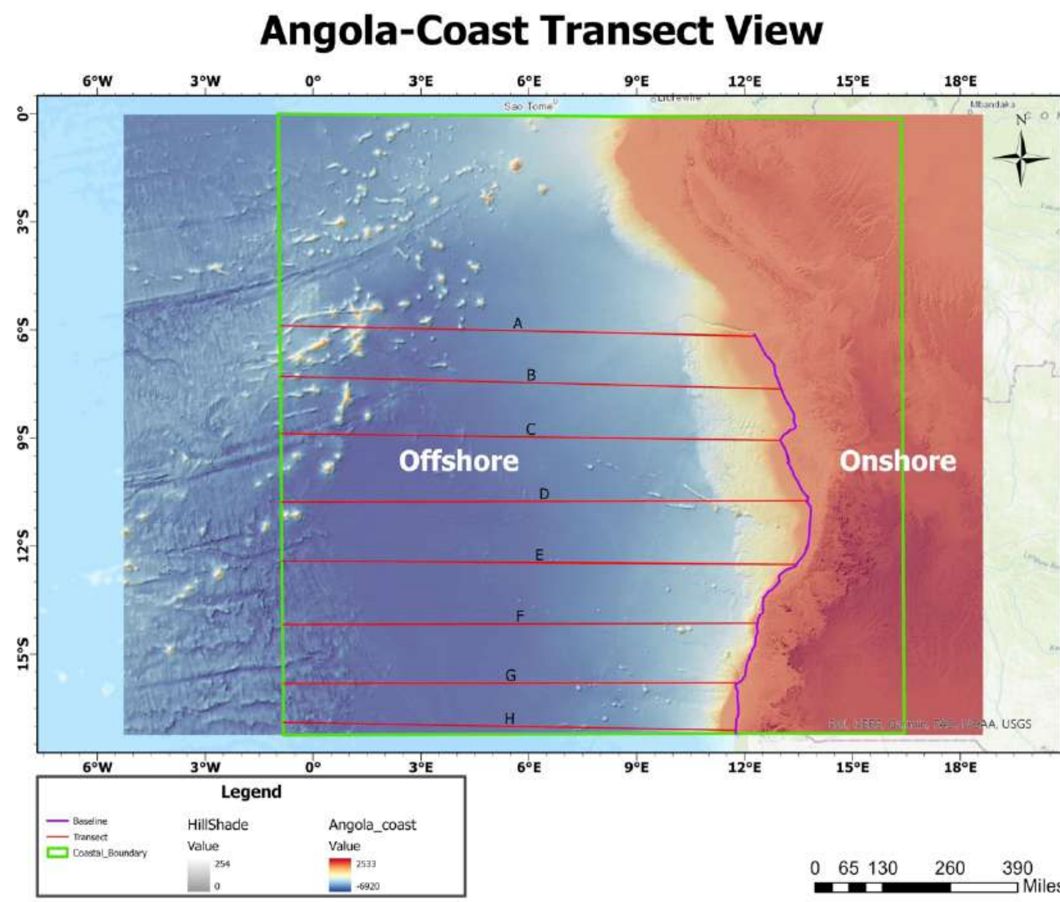
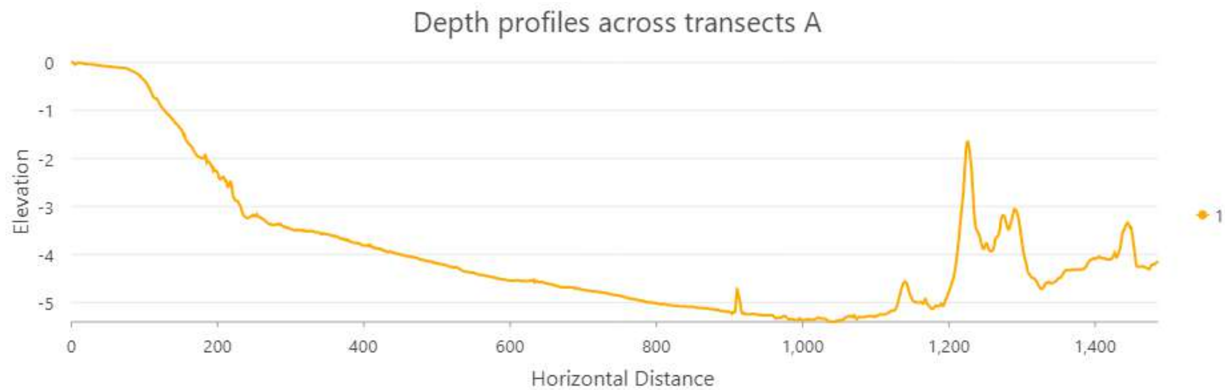


Figure 4.2: Transect View Map showing profiles A–H.

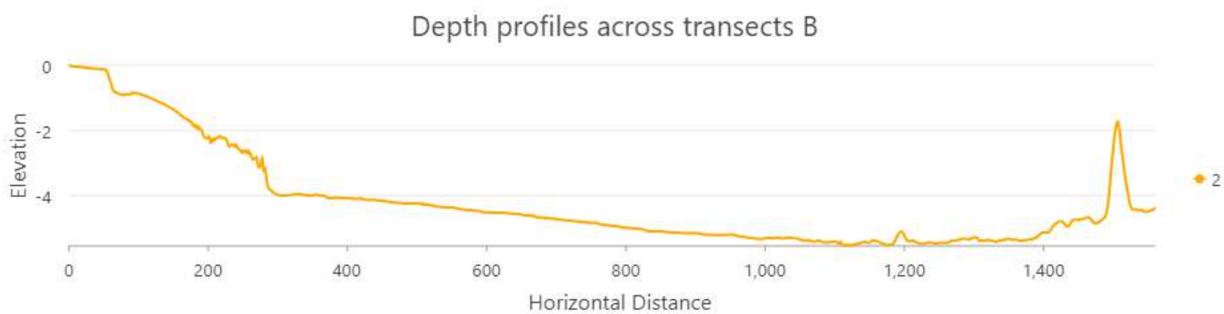
4.3 Depth Profiles (A–H)

The depth profiles (Figures 4.3a–h) provided a more quantitative representation of seafloor variation. Each profile illustrated how water depth increased with distance from the shoreline.

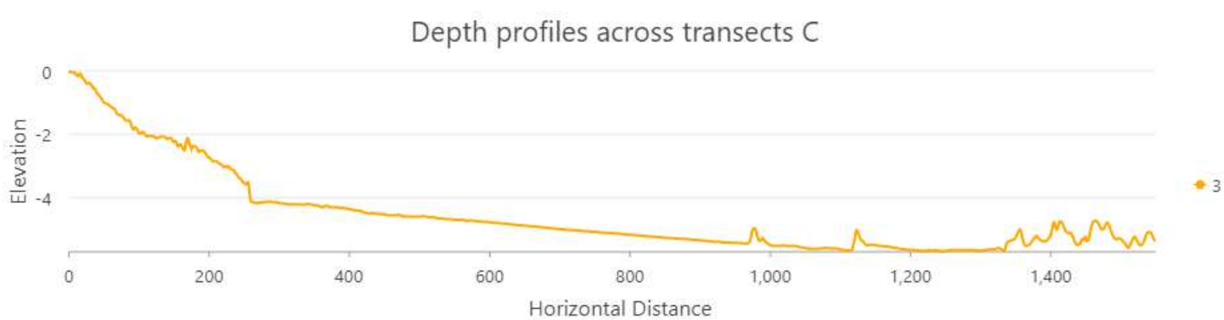
- **Profiles A–C (Northern Angola):** The seabed slopes gradually, with depth increasing slowly even at 30–40 km offshore. This indicates a broad continental shelf that can act as a buffer against wave energy, reducing coastal erosion. These areas are also more favorable for shallow-water ecosystems and small-scale fishing activities.



Figures 4.3a *Depth Profiles across transect A*

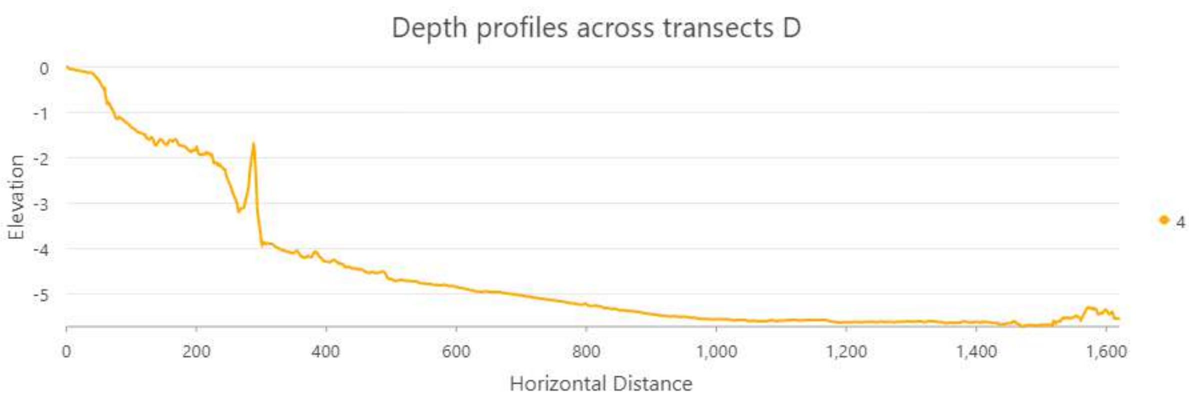


Figures 4.3b *Depth Profiles across transect B*

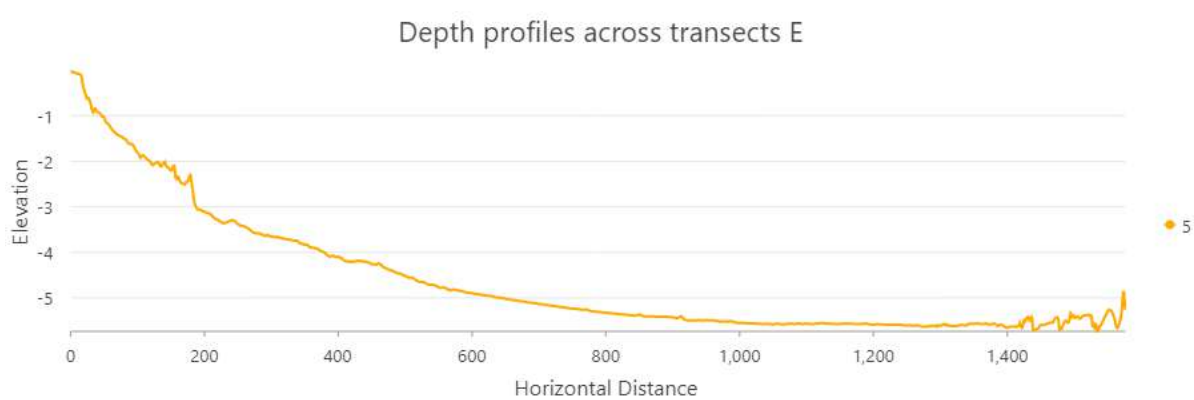


Figures 4.3c *Depth Profiles across transect C*

- **Profiles D–E (Central Angola):** The seabed descends steeply within 10–15 km from the coast, showing depths of several hundred meters. These profiles suggest the presence of submarine canyons that channel sediments directly into deep waters. Such steep slopes increase the vulnerability of this coastline to erosion because wave energy is not dissipated gradually but instead concentrated at the shoreline.

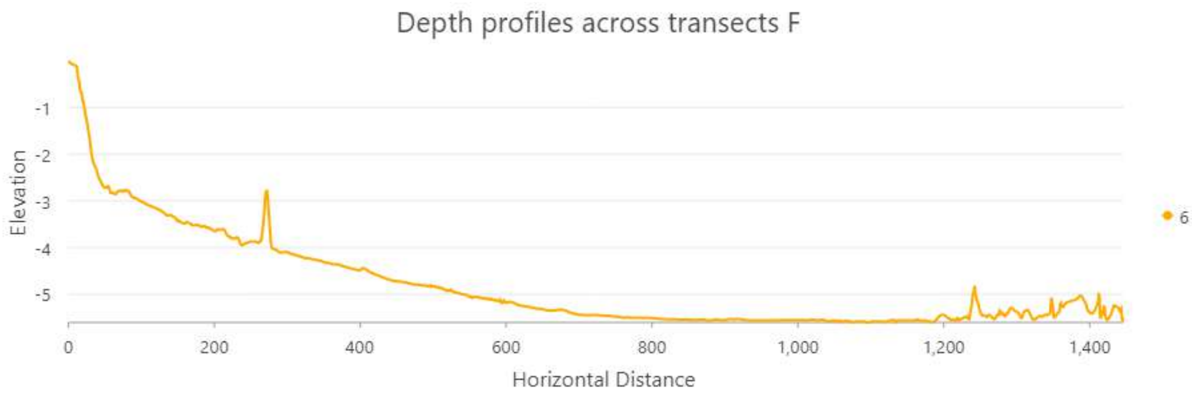


Figures 4.3d *Depth Profiles across transect D*

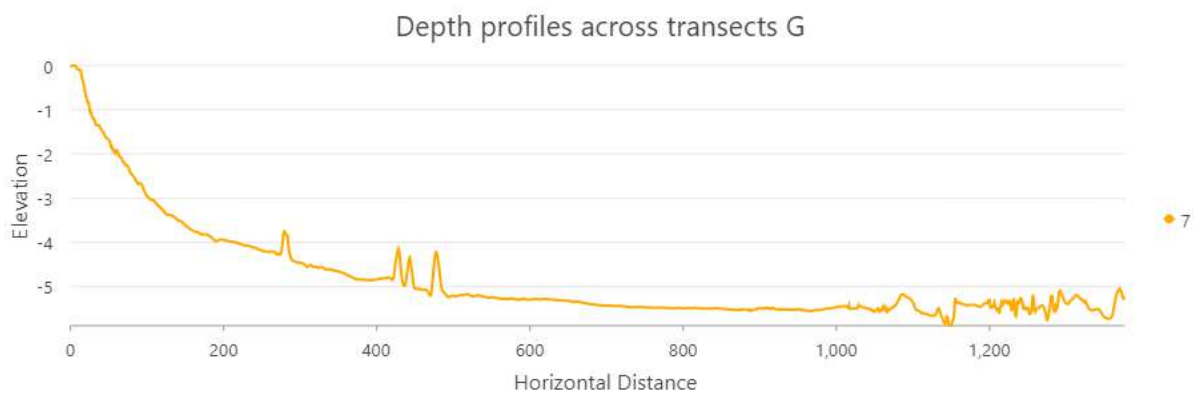


Figures 4.3e *Depth Profiles across transect E*

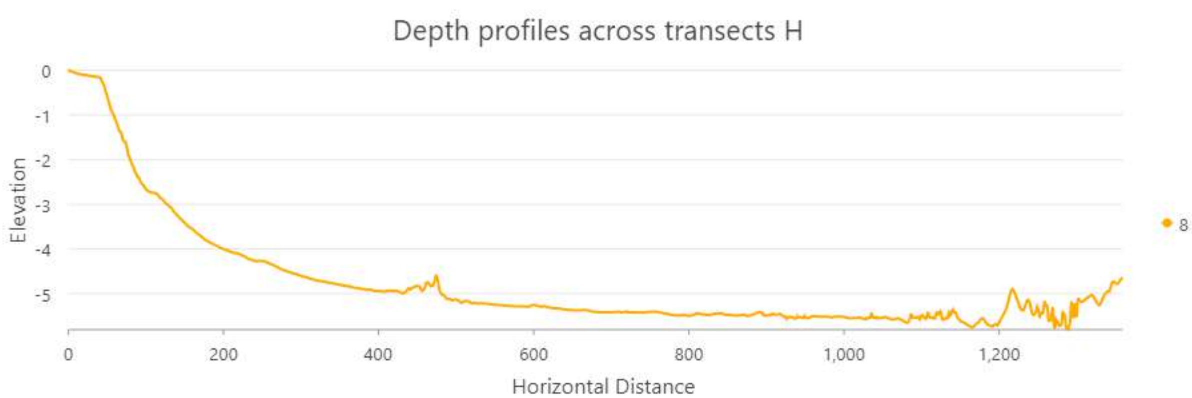
- **Profiles F–H (Southern Angola):** The slope patterns are mixed, with some transects showing gentle descents and others revealing abrupt breaks in slope. This variability reflects the influence of the Benguela upwelling system, which transports sediments and shapes underwater ridges and troughs.



Figures 4.3f *Depth Profiles across transect F*



Figures 4.3g *Depth Profiles across transect G*



Figures 4.3h *Depth Profiles across transect H*

4.4 Composite Depth Profile

When the eight profiles were combined into a single composite plot (Figure 4.4), a broader picture of depth variation along Angola's coast emerged.

- The **northern profiles** are clustered in the shallow range, confirming the presence of wide continental shelves.
- The **central profiles** stand out with sharp drops, highlighting narrow shelves and deep offshore zones.
- The **southern profiles** fall in between, showing intermediate slopes but more irregular patterns, likely shaped by oceanographic currents and sediment movements.

This composite view reinforces the understanding that Angola's seabed is not uniform but instead reflects a transition from depositional to erosional coastal environments.

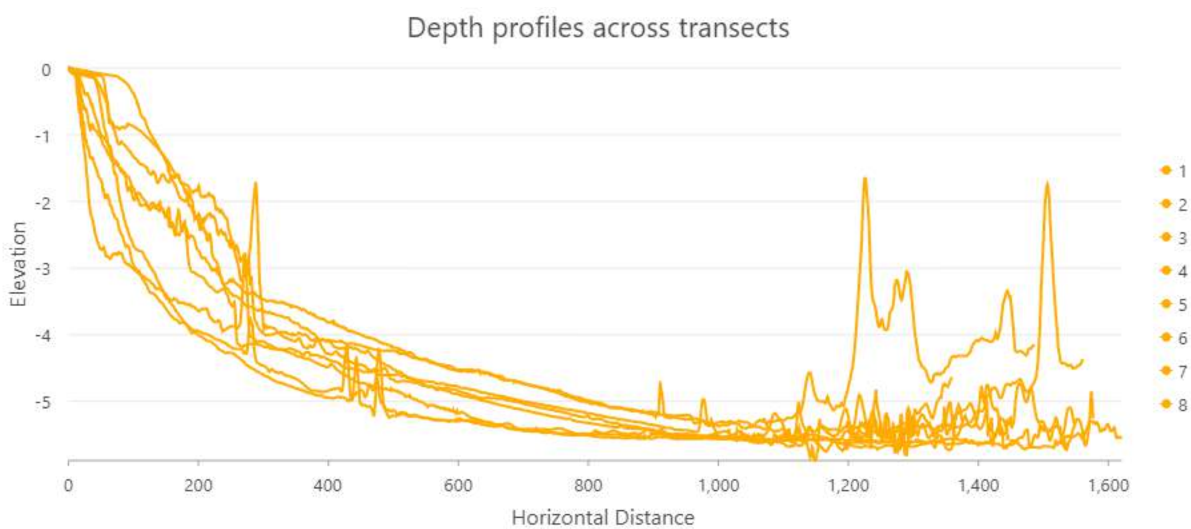


Figure 4.4: Composite Depth Profile.

4.5 Identification of Coastal Features and Habitat Zones

The classification of Angola's coastal terrain and offshore bathymetry (Figures 4.5a and 4.5b) provides a framework for identifying distinct ecological zones along the coast. These zones correspond to slope gradients and geomorphological settings, each supporting unique habitats and ecological functions.

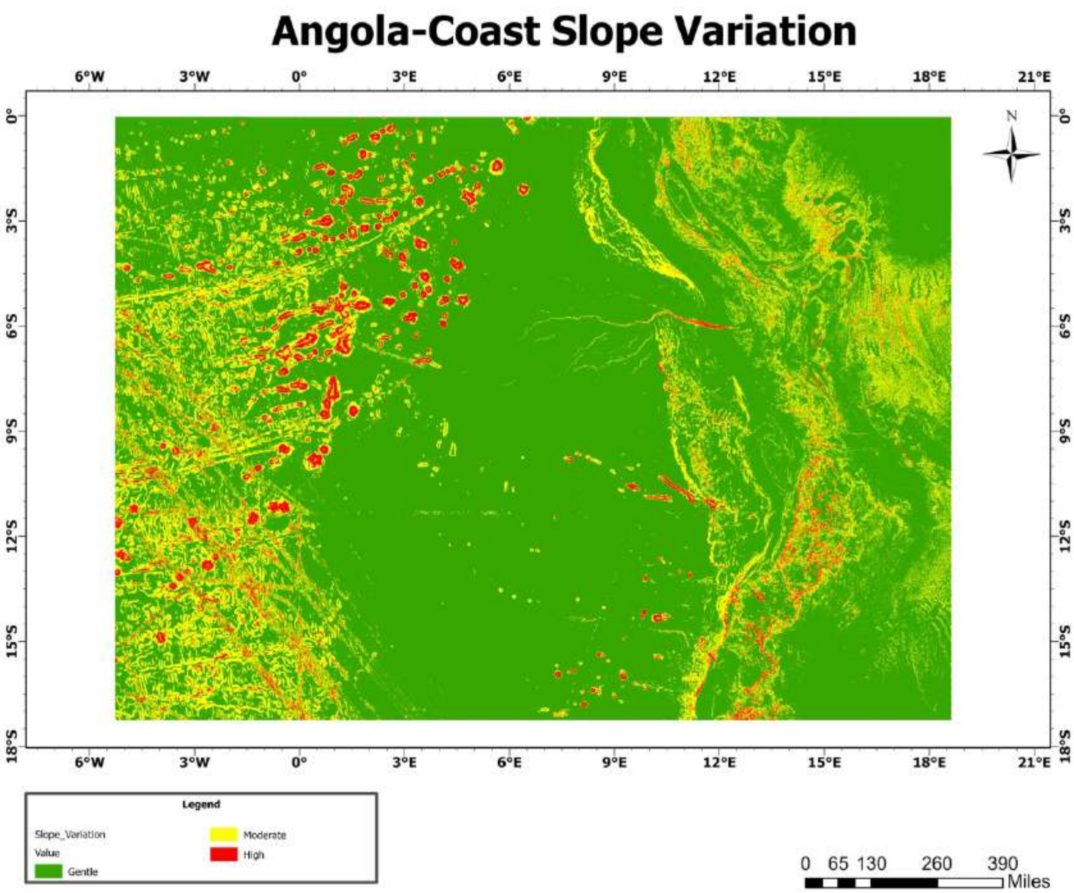


Figure 4.5a: *Slope variation map of the Angola Coast showing gentle, moderate, and high slope gradients.*

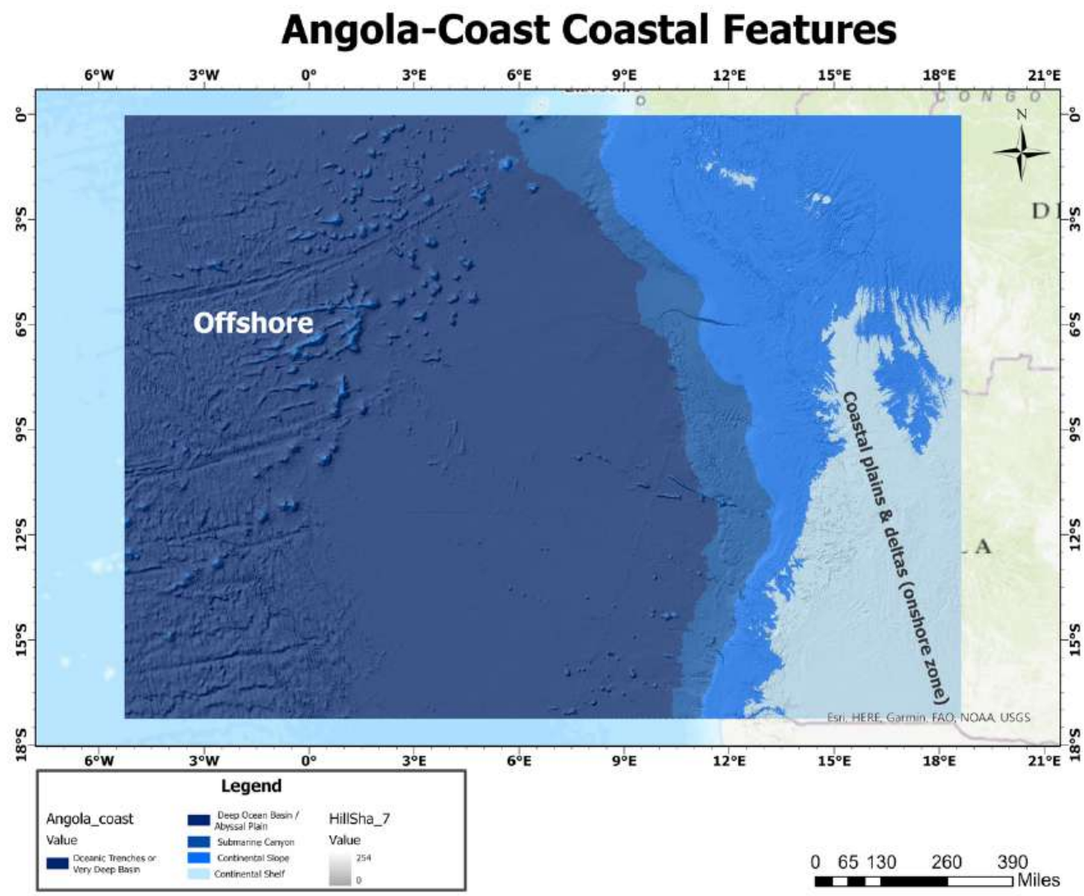


Figure 4.5b: *Bathymetric and geomorphological features of the Angola Coast, including coastal plains, continental shelf, slope, submarine canyons, abyssal plain, and very deep basins.*

1. Coastal Plains and Deltas (Onshore – Nearshore Zone)

As illustrated in the slope variation map (Figure 4.5a), the low-lying coastal plains and deltaic regions dominate the nearshore environment.

- **Habitats include:** mangroves, estuaries, lagoons, and deltaic wetlands.
- **Ecological role:** These areas serve as vital nursery grounds for fish and crustaceans, support migratory birds, and maintain high productivity due to nutrient inflow from rivers.

2. Continental Shelf (Gentle Slopes – Light Blue Zone)

The continental shelf (Figure 4.5b) represents the shallow offshore zone with gentle slopes and relatively high light penetration.

- **Habitats include:** seagrass beds, sandy and muddy seabeds, and localized coral reefs.
- **Ecological role:** This zone forms Angola's richest fishing grounds, sustaining pelagic and demersal species. It also supports benthic organisms such as mollusks, shrimp, and crabs, making it one of the most productive marine zones.

3. Continental Slope (Moderate to Steep Gradient)

Beyond the shelf, the continental slope marks a transition from shallow to deep ocean zones (Figure 4.5a).

- **Habitats include:** cold-water coral reefs, sponge fields, and sediment slope communities.
- **Ecological role:** This zone provides important habitats for migratory species and serves as a corridor linking shallow and deep ecosystems. However, it is also sensitive to submarine landslides and sediment transport processes.

4. Submarine Canyons and Escarpments (Finger-like Features Offshore)

Submarine canyons, visible in the coastal features map (Figure 4.5b), are deep erosional features extending from the shelf to deeper basins.

- **Habitats include:** rocky canyon walls and sediment-rich canyon floors.
- **Ecological role:** Canyons act as funnels that transport organic matter and nutrients into deep waters. They are hotspots of biodiversity, supporting deep-sea corals, crustaceans, and demersal fish.

5. Abyssal Plain (Flat Deep Ocean Floor)

Farther offshore, the abyssal plain (Figure 4.5b) represents the vast, flat, deep-sea floor.

- **Habitats include:** soft-sediment communities (polychaetes, sea cucumbers, echinoderms) and isolated seamount ecosystems.
- **Ecological role:** Although low in productivity, this zone supports specialized fauna and plays an essential role in long-term carbon storage through sediment deposition.

6. Very Deep Basins and Trenches (>4,800 m)

The deepest offshore features include very deep basins and potential trench systems, which represent extreme marine environments.

- **Habitats include:** hadal trench ecosystems (amphipods, snailfish, extremophiles) and chemosynthetic communities near hydrothermal vents (if present).
- **Ecological role:** These are unique ecosystems with highly adapted species, important for studying evolutionary processes and biodiversity in extreme environments.

The slope variation and coastal bathymetry maps (Figures 4.5a and 4.5b) demonstrate how Angola's coastline transitions from **nutrient-rich deltas and productive shelves** to **steep slopes and submarine canyons**, before flattening into the **deep abyssal plain and trenches**. Each zone sustains distinct habitats that provide essential ecological services — from supporting artisanal fisheries nearshore to hosting unique deep-sea biodiversity offshore.

4.6 3D Visualization of Coastal Region

The 3D visualization (Figure 4.6) transformed the bathymetric data into a more realistic model of the seafloor. This product highlighted slope breaks, underwater ridges, and depressions more clearly than the 2D hillshade.

- The **northern coast** appeared as a gently sloping plane, confirming the shelf-dominated environment.
- The **central coast** showed dramatic slope changes, visualized as steep escarpments plunging into deeper waters.
- The **southern coast** revealed alternating ridges and troughs consistent with sediment redistribution by strong ocean currents.

Such visualizations are particularly useful for decision-makers and non-technical audiences, as they provide an intuitive grasp of the seabed's complexity.

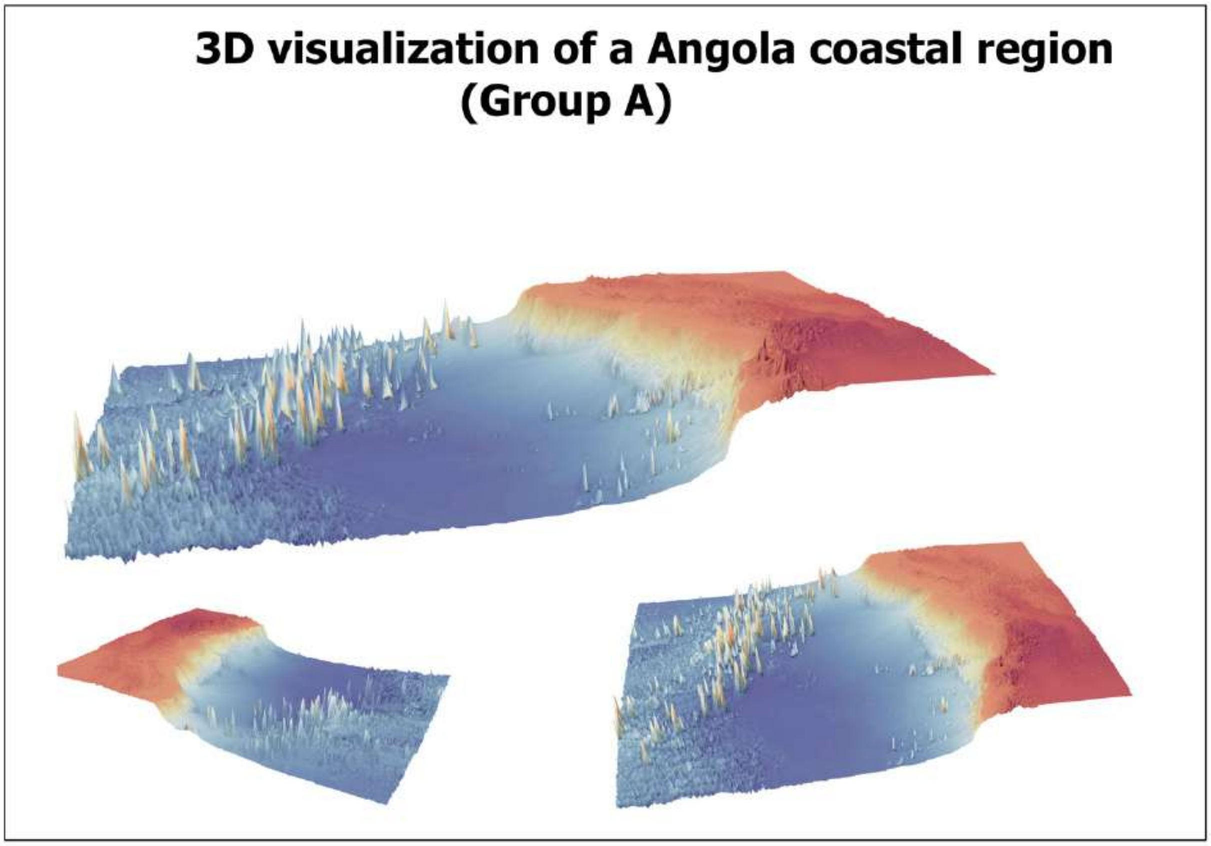


Figure 4.6: 3D Visualization of Angola Seafloor.

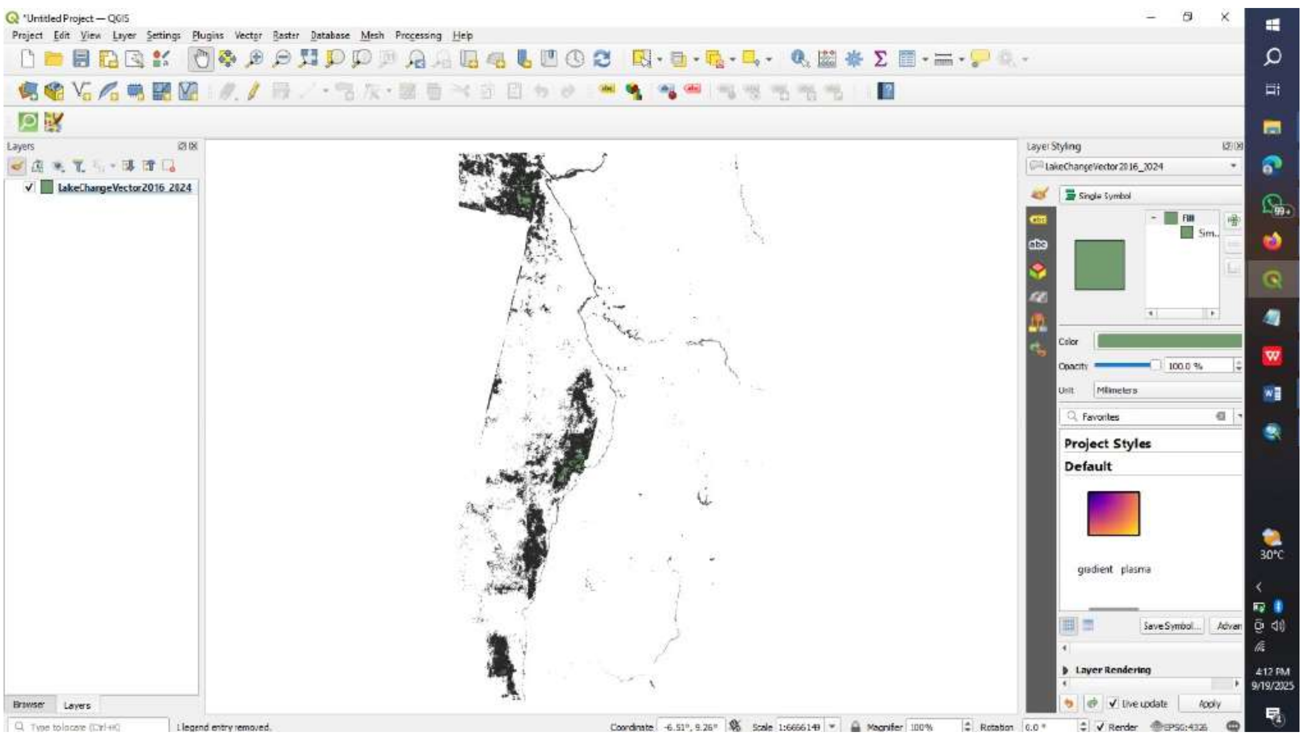
4.7 Shoreline Change Detection Results

The shoreline analysis using DSAS revealed important trends between 2000 and 2023 (Figure 4.7).

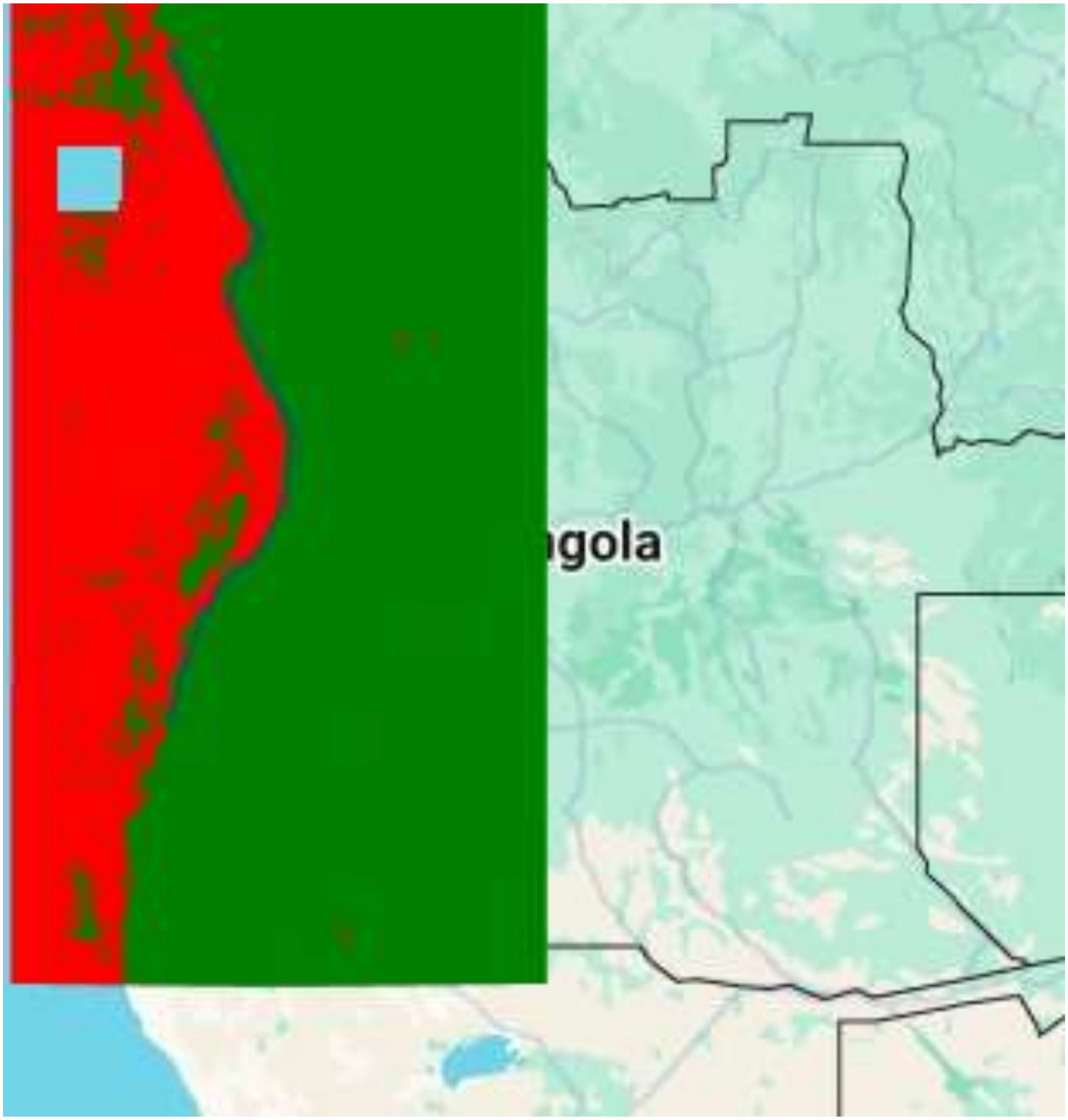
- **Erosion hotspots:** Central Angola showed consistent shoreline retreat, with erosion rates reaching several meters per year. This is consistent with the steep seabed slopes observed in the bathymetric profiles, which reduce sediment retention.
- **Accretion zones:** Northern Angola displayed localized accretion, especially near river mouths and estuaries. This reflects the role of rivers in supplying sediments that are deposited along the shallow shelves.

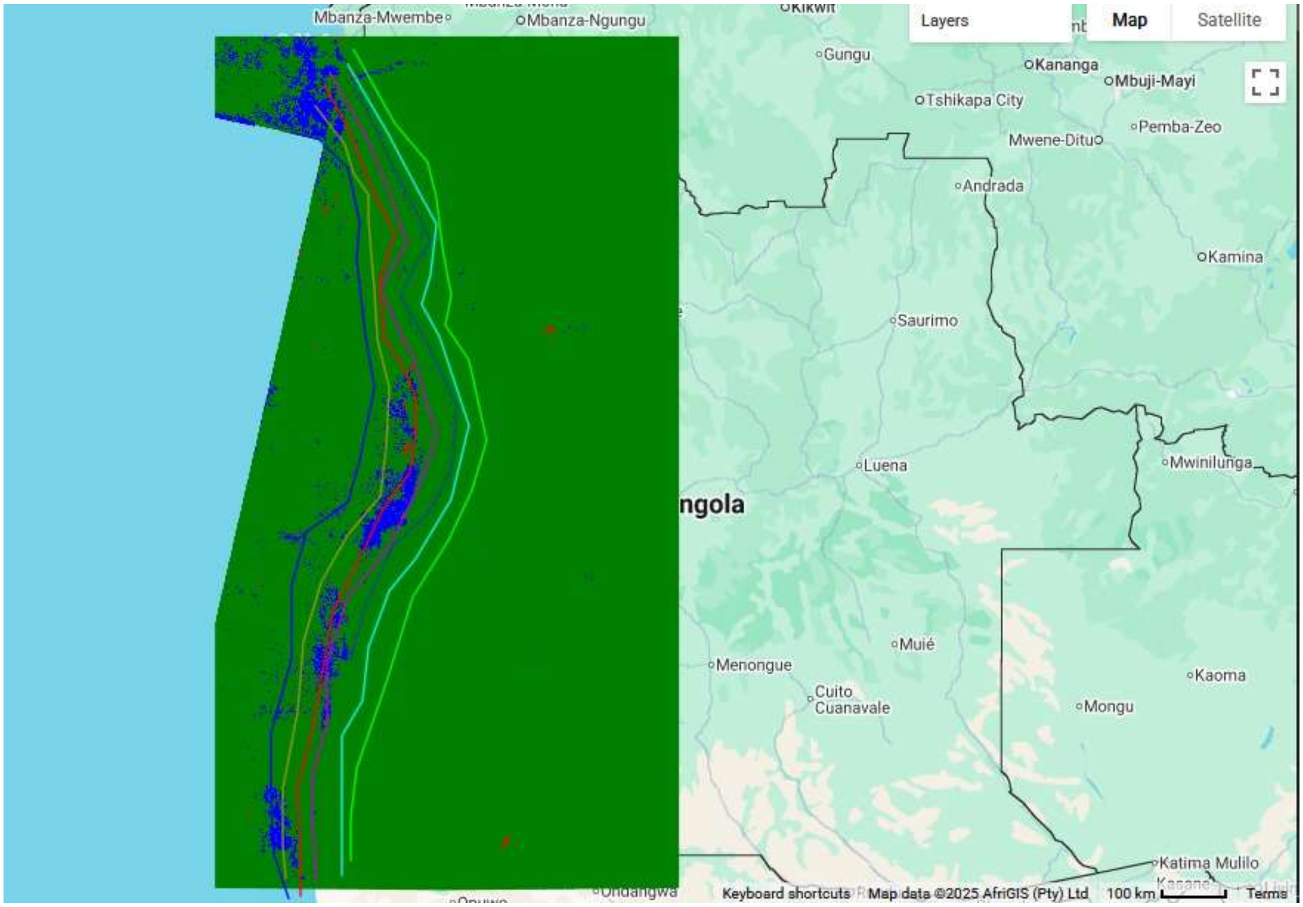
- **Temporal changes:** Erosion was most severe between 2000 and 2010, but recent images show some stabilization, possibly linked to natural sediment replenishment or human interventions such as groynes or harbor works.
- **Southern coast:** Shoreline changes were mixed, with both erosion and accretion occurring in different stretches, influenced by the Benguela Current's complex sediment transport processes.

These results highlight the dynamic balance between erosion and deposition along Angola's coast.



transect_id	year	distance_m	change_m	rate_m_per_year
0	2018	120.5		
0	2024	108.2	-12.3	-2.05
1	2018	115.0		
1	2024	110.0	-5.0	-0.83
2	2018	130.3		
2	2024	135.7	5.4	0.90
3	2018	125.8		
3	2024	118.4	-7.4	-1.23
4	2018	140.2		
4	2024	145.0	4.8	0.80
5	2018	110.7		
5	2024	103.1	-7.6	-1.27
6	2018	128.9		
6	2024	132.5	3.6	0.60
7	2018	135.4		
7	2024	-9999	-9999	-9999
8	2018	122.6		
8	2024	117.8	-4.8	-0.80
9	2018	118.3		
9	2024	123.9	5.6	0.93


gola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Mwene-Ditu
Andrada
Kamina
Saurimo
Luena
Mwinilunga
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Menongue
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Map data ©2025 AfriGIS (Pty) Ltd
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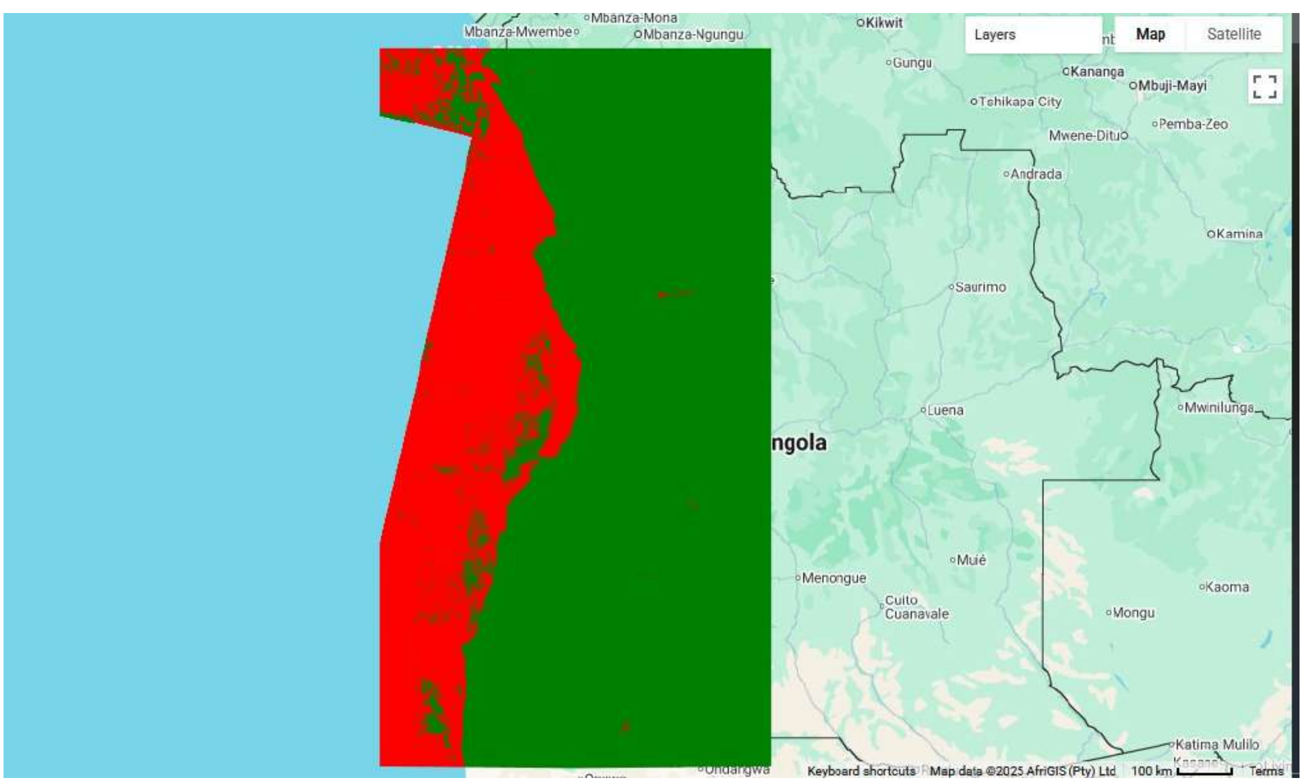
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Map data ©2025 AfriGIS (Pty) Ltd
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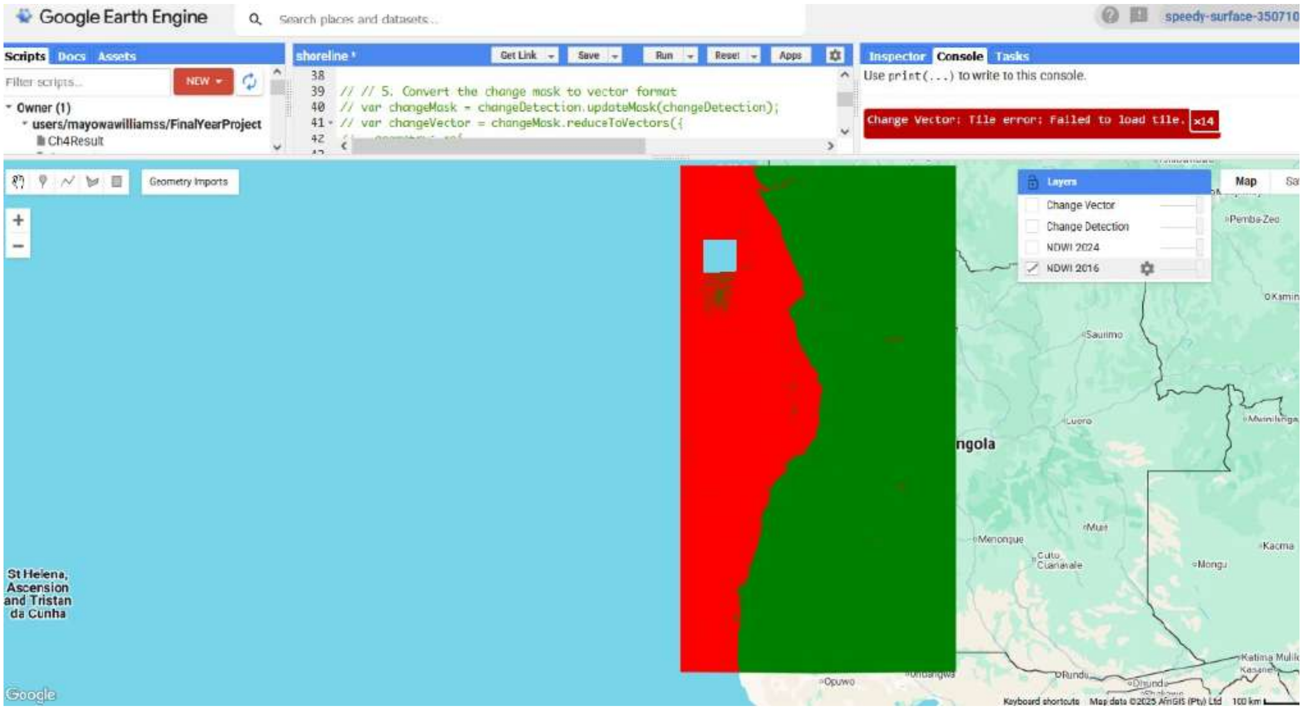

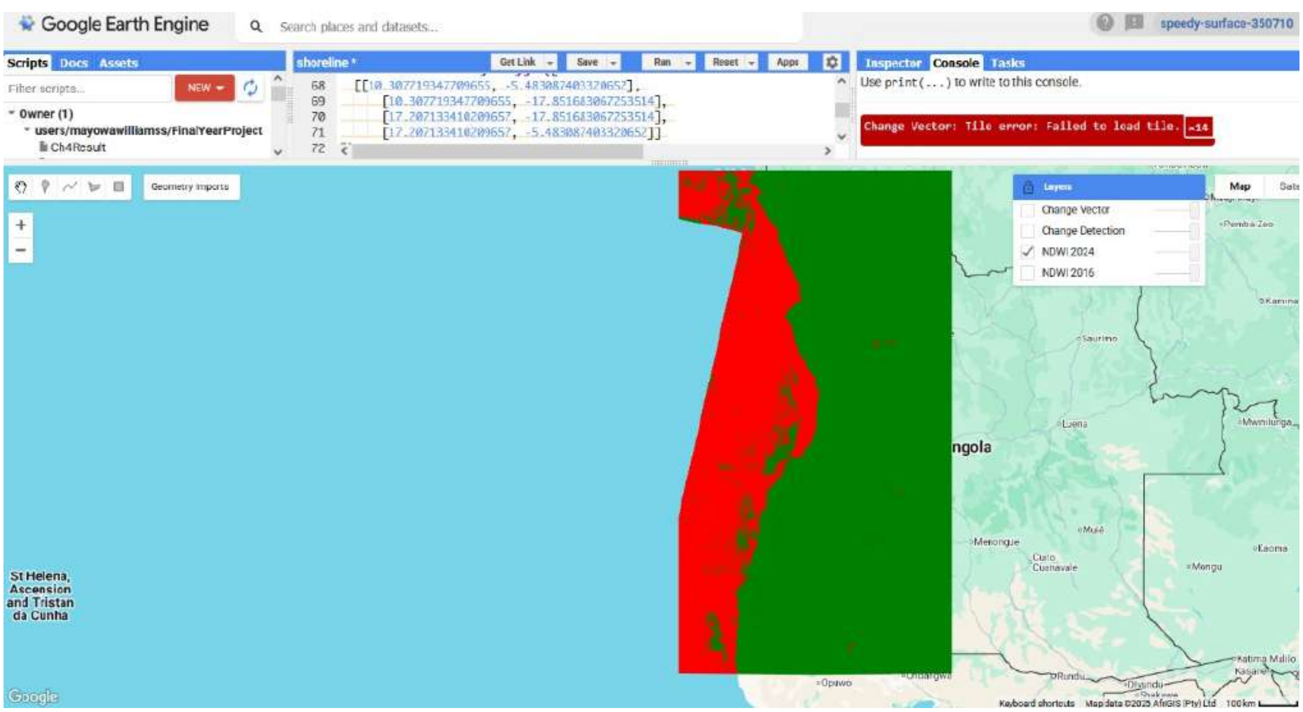

Figure 4.7: Shoreline Change Detection Map (DSAS), NDWI 2024, GEE legend

4.8 Interpretation and Implications

Bringing together the bathymetric and shoreline analyses provides a holistic understanding of coastal dynamics in Angola:

- **North Angola:** Characterized by wide shelves and depositional environments, this region is important for biodiversity conservation. However, its low elevation makes it highly vulnerable to sea-level rise and storm surges.
- **Central Angola:** Defined by steep seabeds and erosion-prone shorelines, this region faces the greatest risk of land loss. Coastal protection strategies are urgently needed here.
- **South Angola:** Marked by dynamic sediment redistribution under the Benguela Current, this area supports rich fisheries but faces challenges of shifting sandbanks and unstable coastlines.

The findings imply that **management strategies must be region-specific**:

- In the north, focus on protecting habitats and managing flooding risks.
- In the central region, invest in erosion control and shoreline stabilization.
- In the south, balance fisheries development with monitoring of sediment dynamics.

Ultimately, this integrated approach demonstrates how combining bathymetry with shoreline change analysis provides both scientific insights and practical tools for coastal management.

CHAPTER 5: CONCLUSION AND RECOMMENDATIONS

5.1 Summary of Findings

This study applied Remote Sensing (RS) and Geographic Information Systems (GIS) techniques to analyze spatio-temporal changes along the Angolan coastline through bathymetric assessment and shoreline change detection. Bathymetric products such as hillshade maps, depth profiles, slope maps, and 3D visualizations revealed significant variability in seafloor morphology across the northern, central, and southern sectors. The northern coast is characterized by wide continental shelves and depositional environments, providing biodiversity hotspots but facing risks from sea-level rise and flooding. The central coast displayed narrow shelves with steep slopes and submarine canyons, making it highly prone to erosion. The southern coast exhibited dynamic geomorphology shaped by the Benguela Current, with mixed erosion and accretion trends influencing sediment transport and fisheries productivity.

Shoreline change detection using DSAS highlighted active erosion in central Angola, localized accretion in the north near river mouths, and mixed dynamics in the south. The combined analysis provided a holistic understanding of Angola's coastal processes, identifying critical areas for conservation, risk management, and sustainable development planning.

5.2 Recommendations

Based on the findings, the following recommendations are proposed:

1. **Region-specific coastal management strategies**:
   - **Northern Angola**: Prioritize habitat conservation (mangroves, estuaries) and flood risk management.
   - **Central Angola**: Implement shoreline stabilization measures such as groynes, seawalls, and managed retreat strategies to combat severe erosion.
   - **Southern Angola**: Strengthen fisheries management and monitor sediment redistribution under the Benguela Current.
2. **Integration of geospatial technologies into national coastal policies**: Promote continuous use of RS, GIS, and open-source datasets (GEBCO, GRID3, satellite imagery) for decision-making.
3. **Erosion control and adaptation measures**: Establish early warning systems and erosion monitoring networks, especially in central Angola where vulnerability is highest.
4. **Sustainable development planning**: Balance industrial expansion (ports, oil facilities, tourism) with environmental conservation to reduce long-term risks.
5. **Capacity building and collaboration**: Train local researchers and policymakers in geospatial applications and encourage regional cooperation for coastal monitoring.

5.3 Suggestions for Future Work

Future studies should:

- Incorporate **higher-resolution bathymetric and satellite datasets** (e.g., LiDAR, UAV-based imagery) to improve shoreline mapping accuracy.
- Extend temporal coverage by including **longer historical datasets** for a deeper understanding of long-term coastal evolution.
- Explore the **impacts of climate change**, particularly sea-level rise and intensifying storm events, on Angola's coastline.
- Assess the **socioeconomic implications** of coastal change, focusing on vulnerable communities, infrastructure, and livelihoods.
- Develop **predictive models** that integrate bathymetric, hydrodynamic, and climate data to forecast future shoreline positions and erosion risks.

By addressing these areas, future research will enhance the capacity for evidence-based coastal management and ensure the sustainable use of Angola's coastal resources.